

Observation: If d is not a square, then

$$N(\alpha) = 0 \Leftrightarrow \alpha = 0.$$

$$\alpha = x + y\sqrt{d} = 0 \Rightarrow x = -y\sqrt{d}$$

$$\Rightarrow x^2 = y^2 d$$

$$\Rightarrow N(\alpha) = x^2 - y^2 d = 0.$$

$$N(\alpha) = 0 \Rightarrow x^2 - y^2 d = 0 \Rightarrow x^2 = y^2 d$$